INTRODUCTION TO COMPUTING SCIENCES

LECTURE 1: Basic Concepts • Data and Information • Hardware and Software • Computer Networks

About these notes: The original lecture points supplied are retained and expanded with introductory explanations, examples and diagrams. Extra explanations are study support; follow your lecturer's instructions if a class definition or emphasis differs.

Learning outcomes

• Explain what computing and a computer are.

• Distinguish between data and information and give examples.

• Identify computer hardware and explain the two main software categories.

• Define a computer network and describe its purposes.

1. Basic Concepts of Computing

Computing

Computing is the use of computers and related technologies to collect, process, store, retrieve and communicate information. Computing is used in education, banking, healthcare, business, communication and entertainment.

Computer

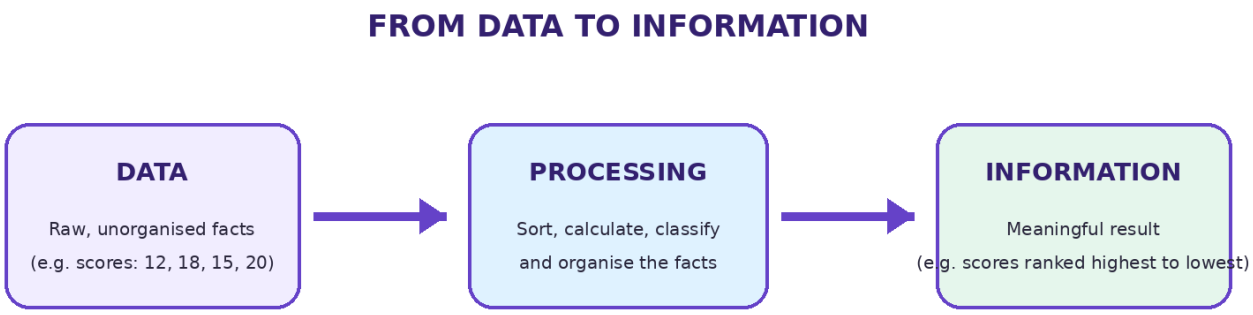
The lecture notes define a computer as an electronic device that accepts data and provides output. In expanded terms, a computer accepts input, processes it according to instructions, can store data and results, and produces output that is useful to the user.

Examples of computers

Desktop computers, laptops, tablets, smartphones, servers and some embedded devices such as the computer systems in vehicles.

The basic information-processing cycle

Input: data and instructions are entered. Processing: the computer works on the data. Output: results are presented. Storage: data and results can be saved for later use.



2. Data and Information

Data

Data are raw, unprocessed or unorganised facts. They may be numbers, words, measurements, symbols, images or sounds. By themselves, individual data items may have little meaning.

Information

Information is data that has been processed, organised or interpreted so that it is meaningful and useful to someone.

From data to information

Data becomes information when it is organised or processed for a purpose. The result depends on the question being asked and the way the data is arranged.

DATA (raw facts)	INFORMATION (organised result)
List of students in a class	Students organised according to attendance
Random numbers in a set	Numbers arranged in order or grouped by value
Daily temperature readings	Average temperature for the week

3. Hardware and Software

Hardware

Hardware refers to the physical parts of a computer system that can be seen and touched. Examples include the monitor, keyboard, mouse, system unit, processor, memory, storage drive, printer and speakers.

Software

Software is a collection of programs and instructions that tells computer hardware what tasks to perform. Software is not a physical component that can be touched.

System software

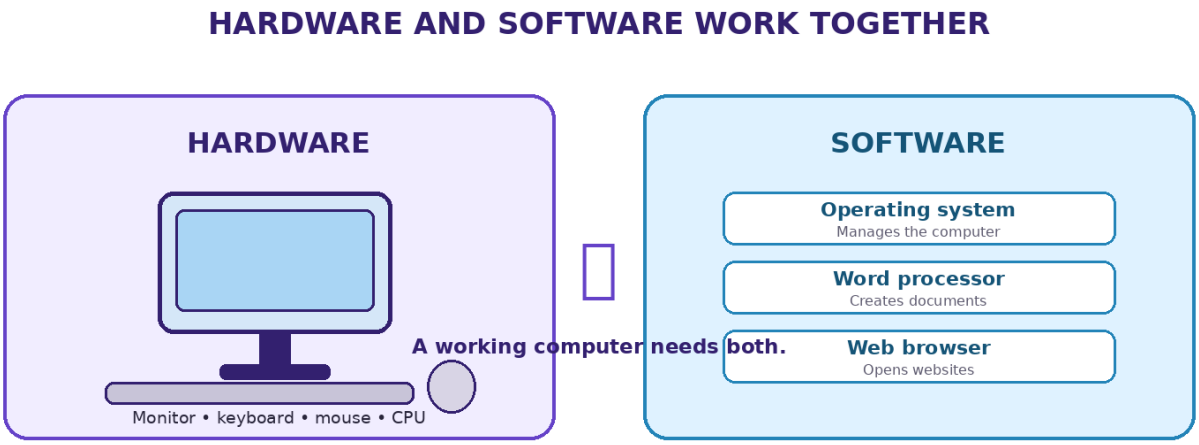
System software manages the computer and provides a platform for other programs. Examples include operating systems such as Windows, macOS, Linux, Android and iOS, as well as device drivers and utility programs.

Application software

Application software helps users perform particular tasks. Examples include word processors, spreadsheets, presentation programs, web browsers, media players and learning applications.

How they work together

Hardware provides the physical equipment; software provides the instructions. For example, a word-processing application uses the computer's processor, memory, screen and storage to create and save a document.



4. Computer Networks

Definition

A computer network is a set of interconnected computers and other devices that can communicate with one another. This follows the definition in the supplied lecture notes.

Main purposes

Networks allow connected devices to share resources, exchange information and communicate. Shared resources may include files, printers, applications and internet connections.

Examples of networked devices

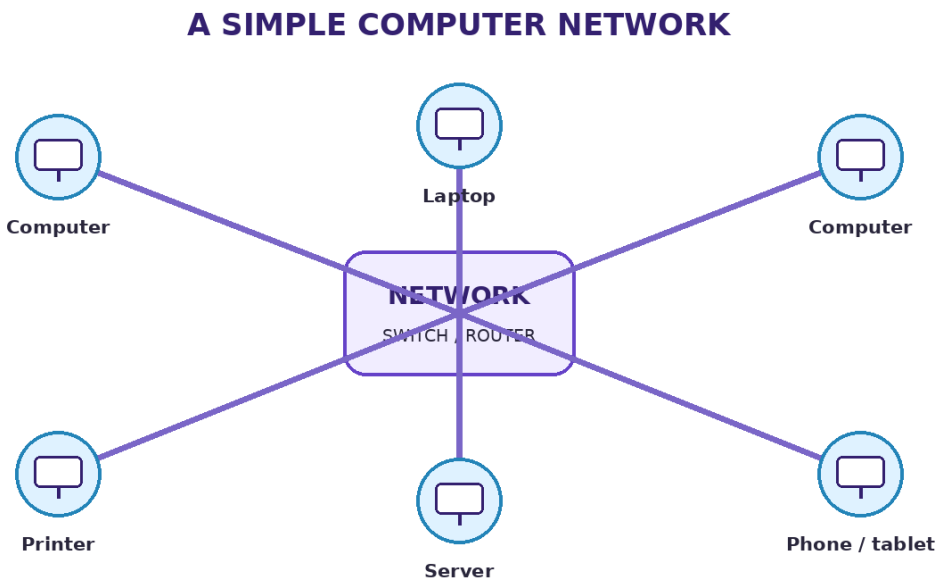
Desktop computers, laptops, smartphones, tablets, servers, printers and network equipment such as routers and switches.

Common network categories

A Local Area Network (LAN) connects devices within a limited area, such as a home, laboratory, office or school building. A Wide Area Network (WAN) covers a much larger geographical area and can connect networks across cities or countries. The internet is the best-known example of a global network of networks.

Everyday examples

Students may use a school network to access shared files, use a shared printer, visit learning platforms or communicate with other users.



Quick revision summary

• A computer accepts data, processes it according to instructions and produces output.

• Data means raw facts; information is data organised into a useful form.

• Hardware consists of physical components; software consists of programs and instructions.

• System software manages computer operations; application software helps users complete specific tasks.

• A computer network connects devices so they can share resources, exchange information and communicate.

Check your understanding

1. Define a computer in your own words.

2. What is data? How is it different from information?

3. Give two examples of data being organised into information.

4. State four examples of computer hardware.

5. Explain the difference between system software and application software, giving two examples of each.

6. What is a computer network?

7. List three purposes of a computer network.

8. What is one difference between a LAN and a WAN?

Key terms

Computer: An electronic device that accepts and processes data to produce output.

Data: Raw, unprocessed facts.

Information: Processed or organised data that is meaningful.

Hardware: The physical components of a computer.

Software: Programs and instructions used by a computer.

Network: Connected devices that communicate and share resources.

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